# MATERIAL SAFETY DATA SHEET

Version：V1.0

# PU-202B

| 1. Identification | |
|---|---|
| **1.1 Product name:** | Waterborne polyurethane dispersion PU-202B |
| **Chemical category:** | Modified polyurethane resin aqueous dispersion |
| **1.2 Product use suggestions and restrictions:** | Used in paper coating, leather finishing, textile coating, plastic coatings and other industries. |
| **1.3 Supplier information:** | |
| **Name of supplier:** | Yingde Guocai New Material Co., Ltd. |
| **Supplier address:** | Kaidi Industrial Park, Genggukeng, Taiping Village, Baisha Town, Yingde, Guangdong, China |
| **Tel:** | 86-763-2811205 |
| **Fax:** | 86-763-2811024 |

## 2.Hazards Identification

**2.1 Classification of the substance or mixture:**

Reproductive toxicity (Category 1B), H360

**2.2 GHS Label Elements:**

**GHS pictograms:**



**Signal word:** Danger

**Hazard statements:**

H360 May damage fertility or the unborn child.

**Precautionary statements:**

**Prevention:**

P201 Obtain special instructions before use.

P202 Do not handle until all safety precautions have been read and understood.

P280 Wear protective gloves/clothing/eye protection/face protection.

## 2.Hazards Identification

**Response:**

P308+P313 IF exposed or concerned: Get medical advice/attention.

**Storage:**

P405 Store locked up.

**Disposal:**

P501 Dispose of contents/container in accordance with local regulations.

**2.3 Other hazards:** Risk of cutaneous absorption of N-Methyl-2-pyrrolidone (NMP).

## 3.Composition/Information on Ingredients

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|---|---|---|
| **3.1 Product type:** | Mixture | |
| **3.2 Ingredients:** | | |
| **Chemical name** | **CAS NO.** | **Concentration % (w/w)** |
| Polyurethane dispersion | 9009-54-5 | 26.0~29.0 |
| Deionized water | 7732-18-5 | 59.0~63.0 |
| N-Methyl-2-pyrrolidone (NMP) | 872-50-4 | 11.0~12.0 |

## 4.First aid measures

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| **4.1 General measures:** | Remove all contaminated clothing immediately. |
| **4.2 Ingestion:** | Do NOT induce vomiting. Seek medical attention immediately. |
| **4.3 Eye contact:** | Hold eyelids open and flush with plenty of warm water for at least 10 minutes. Consult an ophthalmologist. |
| **4.4 Skin contact:** | Wash immediately with plenty of soap and water. If skin reaction occurs, seek medical advice. |
| **4.5 Inhalation:** | If respiratory irritation occurs, seek medical advice. |

## 5.Fire fighting measures

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| **5.1 Suitable extinguishing agents:** | Carbon dioxide ($CO_2$), foam, extinguishing powder. For large fires, use water spray. |
| **5.2 Unsuitable extinguishing agents:** | High-volume water jet. |

| 5.Fire fighting measures | |
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| **5.3 Special hazards of substances or mixtures:** | Combustion may release carbon monoxide, carbon dioxide, nitrogen oxides and trace amounts of hydrogen cyanide.<br>Do not breathe fumes/smoke in case of fire or explosion. |
| **5.4 Firefighting precautions protective equipment:** | Firefighters must wear self-contained breathing apparatus.<br>Do not allow contaminated fire-fighting water to enter soil, groundwater, or surface water. |

| 6.Measures for accidental leakage | |
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| **6.1 Firefighting precautions protective equipment:** | Wear protective equipment.<br>Ensure adequate ventilation. Keep unauthorized persons away. |
| **6.2 Environmental precautions:** | Do not discharge into drains, wastewater, or soil. |
| **6.3 Methods and material for containment and cleaning up:** | Absorb with chemical absorbent material or dry sand if necessary, and store in tightly closed containers. |

| 7.Operation and storage | |
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| **7.1 Precautions for safe handling:** | Observe common chemical precautions during handling. Avoid contact with skin and eyes.<br>Keep away from food, drink and tobacco. Wash hands before breaks and at the end of workday. Store work clothing separately.<br>Change contaminated or soaked clothing. |
| **7.2 Conditions for safe storage:** | Keep containers tightly closed in a dry, well-ventilated place. To maintain product quality, strictly follow the storage conditions in our technical data sheet. |

| 8.Exposure control / personal protection | |
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| **8.1 Exposure controls:** | |
| **Respiratory protection:** | Respiratory protection equipment is required during spraying. |
| **Hand protection:** | Protective gloves are recommended. |
| **Suitable material for protective gloves:** | EN 374-3 |

| 8.Exposure control / personal protection | |
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| **Fluororubber – FKM:** | Thickness ≥ 0.4 mm; Breakthrough time ≥ 480 min. |
| **Butyl rubber – IIR:** | Thickness ≥ 0.5 mm; Breakthrough time ≥ 480 min. |
| **Nitrile rubber – NBR:** | Thickness ≥ 0.35 mm; Breakthrough time ≥ 480 min. |
| **Recommendation:** | Contaminated gloves should be discarded. |
| **Eye protection:** | Wear protective goggles or face shield. |
| **Body protection:** | Wear appropriate protective clothing. |
| **8.2 Engineering controls:** | No exposure limit information is available under EC Directive 2006/121/EC. |

| 9.Physical and chemical properties | |
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| **9.1 Appearance:** | Transparent liquid |
| **9.2 Odour threshold:** | No data available. |
| **9.3 pH value:** | 7-9 |
| **9.4 Ionicity:** | Anionic |
| **9.5 Initial boiling point:** | Approx. 100℃ |
| **9.6 Flash point:** | Not applicable |
| **9.7 Evaporation rate:** | No data available. |
| **9.8 Flammability:** | Not applicable |
| **9.9 Heat of combustion:** | Not applicable |
| **9.10 Saturated vapour pressure:** | No data available. |
| **9.11 Relative vapour density:** | No data available. |
| **9.12 Density:** | 1.0-1.1 g/cm³ |
| **9.13 Solubility in water:** | Miscible |
| **9.14 Surface tension:** | No data available. |
| **9.15 log Pow (n-octanol/water partition coefficient):** | No data available. |
| **9.16 Auto-ignition temperature:** | Not applicable |
| **9.17 Ignition temperature:** | No data available. |
| **9.18 Decomposition temperature:** | No data available. |
| **9.19 Dynamic viscosity:** | < 250mPa.s |

| 9.Physical and chemical properties | |
|---|---|
| **9.20 Explosive properties:** | No data available. |
| **9.21 Dust explosion class:** | No data available. |
| **9.22 Solid content:** | 28±2% |
| **9.23 Other information:** | The above data are not product specifications. Please refer to the Technical Data Sheet for specifications. |

| 10.Stability and reactivity | |
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| **10.1 Chemical stability:** | No decomposition if used according to specifications. |
| **10.2 Decomposition temperature:** | Flammable and harmful gases generated during thermal decomposition. |
| **10.3 Possible hazardous reactions:** | No hazardous reactions when stored and handled correctly. |

| 11.Toxicity information | | |
|---|---|---|
| No toxicological studies were available for this product. | | |
| The following are toxicological data for N-Methyl-2-pyrrolidone (NMP): | | |
| **11.1 Acute toxicity:** | **Oral:** | LD50 / Oral / Rat: 4,150mg/kg |
| | **Inhalation:** | LC50 / Inhalation / 1h / Rat: > 5.1mg/l<br>LC50 / Inhalation / 4h / Rat: > 5.1mg/l |
| | **Dermal:** | LD50 / Oral / Rat: > 5,000mg/kg |
| **11.2 Primary skin irritation:** | Species: Rabbit<br>Classification: Irritating to skin<br>Result: Causes skin irritation | |
| **11.3 Primary eye irritation:** | Species: Rabbit<br>Classification: Irritating to eyes<br>Result: Causes serious eye irritation | |
| **11.4 Sensitization:** | Species: Mouse<br>Classification: Not a skin sensitizer<br>Result: Did not cause sensitization in laboratory animals | |
| **11.5 Mutagenicity:** | No mutagenic effects observed in bacterial or mammalian cell cultures.<br>No mutagenic effects observed in animal studies. | |
| **11.6 Carcinogenicity:** | No carcinogenic effects observed in animal studies. | |

| 11.Toxicity information | | |
|---|---|---|
| **11.7 Reproductive toxicity:** | **Fertility:** | Animal experiments showed risks of impaired fertility only at high doses. |
| | **Teratogenicity:** | Animal experiments showed teratogenic effects. May damage the unborn child. |
| **11.8 Specific target organ toxicity (single/repeated exposure):** | May cause respiratory irritation. | |
| **11.9 Aspiration hazard:** | No data available. | |
| **11.10 Additional information:** | Risk of cutaneous absorption of N-Methyl-2-pyrrolidone (NMP). | |

| 12.Ecological information | |
|---|---|
| No ecotoxicological studies are available for this product. | |
| The following are ecotoxicological data for N-Methyl-2-pyrrolidone (NMP): | |
| **12.1 Ecotoxicity:** | Toxicity to fish:<br>LC50 / 96 h / Oncorhynchus mykiss (rainbow trout): 500mg/l<br>Other:<br>EC50 / 72 h / Algae: ＞500mg/l<br>EC50 / 24 h / Daphnia magna: ＞1,000mg/l |
| **12.2 Persistence and degradability:** | Biodegradability: 73 %, 28 d, readily biodegradable. |
| **12.3 Other adverse effects:** | According to ecotoxicological data, the product is classified as non-hazardous to fish and daphnia. |

| 13.Handling precautions | |
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| Dispose of in accordance with applicable national standards and national or local regulations. | |
| Within the European Union, dispose of in accordance with the applicable provisions of the European Waste Catalogue (EWC). | |
| **Disposal methods:** | Empty containers as completely as possible (e.g., by pouring, scraping, or draining until "drip-dry").<br>They may be sent to an appropriate collection point under existing chemical industry recycling schemes.<br>Recycle containers in accordance with national regulations and |

| 13.Handling precautions | |
|---|---|
| | applicable environmental requirements.<br>Do not discharge waste via wastewater. |

| 14.Transportation information | |
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| **14.1 Road and rail transport:** | Not classified as dangerous goods for land transport, in accordance with ADG regulations. |
| **14.2 Maritime transport:** | Not classified as dangerous goods for sea transport, in accordance with IMDG regulations. |
| **14.3 Air transport:** | Not classified as dangerous goods for air transport, in accordance with IATA regulations. |
| **14.4 Special precautions for user:** | Not regulated as dangerous goods.<br>Temperature must not exceed +40℃ and must not fall below +5℃.<br>Keep away from food, acids and bases. |

| 15.Regulatory information |
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| Relevant safety, health and environmental regulations for the substance or mixture |
| **Other provisions:** |
| **Complies with the following regulations:** |
| Regulations on the Safety Management of Hazardous Chemicals, State Council Decree No. 591 |
| GB/T 16483 Safety data sheet for chemical products - Content and order of sections |
| GB 13690 General rules for classification and hazard communication of chemicals |
| GB 30000.2-29 Rules for classification and labelling of chemicals |
| GB 15258 General rules for preparation of precautionary label for chemicals |

| 16.Other information |
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| To the best of our knowledge, the information provided in this Safety Data Sheet is accurate as of the date of publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal, and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. |